FINSTREET

Team: Excel

PROBLEM UNDERSTANDING AND OBJECTIVE

This project aims to design and implement a fully automated trading system that uses machine learning to generate prediction signals. It converts these signals into systematic trading decisions, executes trades, and assesses performance using a chronological walk-forward backtest.

The focus is not on maximizing raw profits but on creating a realistic, disciplined, and reproducible trading process. It emphasizes signal quality, risk management, and robustness within strict data limits. All training market data is restricted to daily OHLCV data from November 1, 2025, to December 31, 2025.

THE DATA

IRCON.NS was the stock chosen for the competition. The training data was daily OHLCV data from 1 November to 31 December. and was used to predict the next 5 days, i.e. 1 January to 8 January.

METHODOLOGY

FEATURE ENGINEERING

The following features were constructed from OHLCV data:
- Relative Strength Index (RSI) was used as the momentum indicator.
- Average True Range (ATR) was used as the volatility indicator.
- Moving Average Convergence Divergence (MACD) was used as trend indicator
- Volume Relative Strength Index was used as the volume indicator.

The prediction target was set as a binary directional label:
- 1 if the next day’s closing price is higher than the current close
- 0 otherwise

METHODOLOGY

MODEL SELECTION AND JUSTIFICATION

A Logistic Regression classifier was chosen as the main predictive model as the dataset is small (about 40 trading days), making complex models like LSTM and neural networks prone to overfitting.

Open, Close , MACD_history, RSI, ATR and volume RSI were used for training the model.

To enhance numerical stability and probability calibration, the model was trained using a pipeline including feature standardization.

The model predicts the probability that the next trading day’s price movement will be positive.

METHODOLOGY

TRADING STRATEGY

Trading signals were generated from predictions using confidence thresholds:

- Long position if predicted probability is 0.55 or higher
- Short position if predicted probability is 0.45 or lower
- No trade otherwise

Inverse Volatility Position sizing was used for risk management.
Rolling Volatility for 20 days was used for volatility calculation.
Exposure was limited from -1 to -1 .

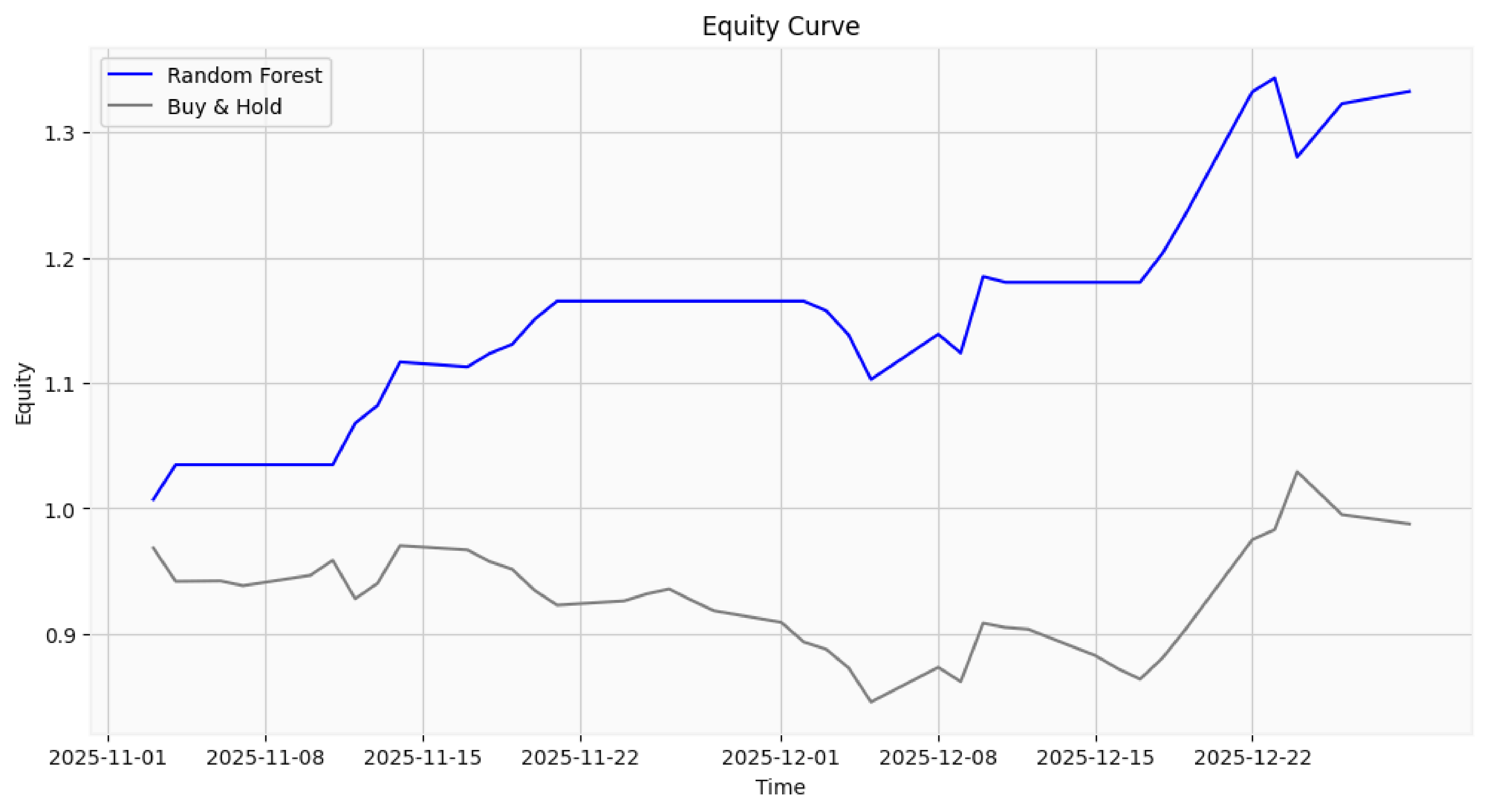
BACKTESTING

The chronological backtest showed promising results indicating the strategy to be consistent and worthy of further testing

Total PnL (%): 33.24

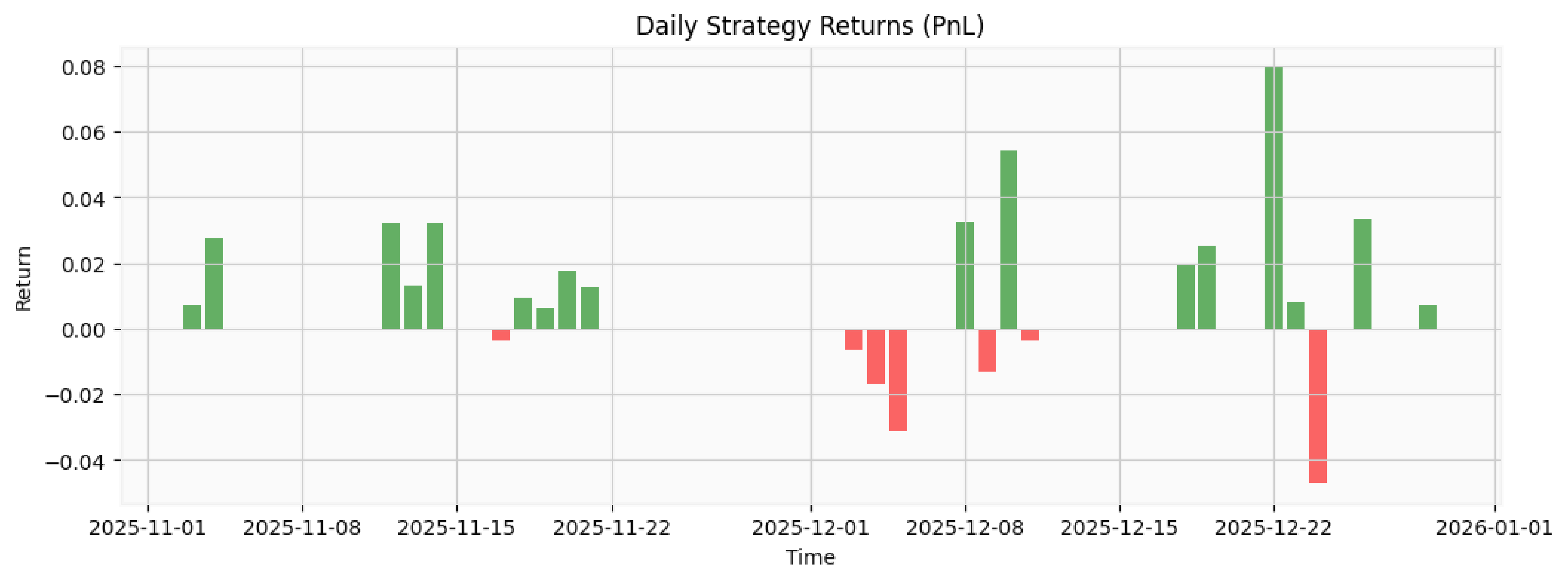
Sharpe Ratio: 5.6

Max Drawdown (%): -5.36



PREDICTIONS

The predictions on the next five days yielded the following results, which are good considering the small training and testing dataset



Total PnL (%): 0.3

Sharpe Ratio: 1.48,

Max Drawdown (%): -0.23

ASSUMPTIONS AND LIMITATIONS

Transaction costs and slippage were considered negligible for simplicity.
The short data window limits the complexity of models that can be trained reliably.
Predictions are directional rather than exact price forecasts.
Despite these limitations, the system exhibits a strong and realistic approach to algorithmic trading under limited conditions.

THANK YOU